

At the eyepiece

What the Eye Still Knows. Thomas J. Watson Fellowship application, Aliaksandr Melnichenka, Berea College.

Everything on these two pages I made myself, with a phone held to an eyepiece or set on a wall. In Belarus the phone was a Redmi Note 7. The 2021 pictures were stacked in Sequator and adjusted in Lightroom. Page 1 is before Berea, page 2 is at Berea, and the people along the route follow on page 3.

Before Berea. Mazyr and the family's country houses, 2020 to 2021.



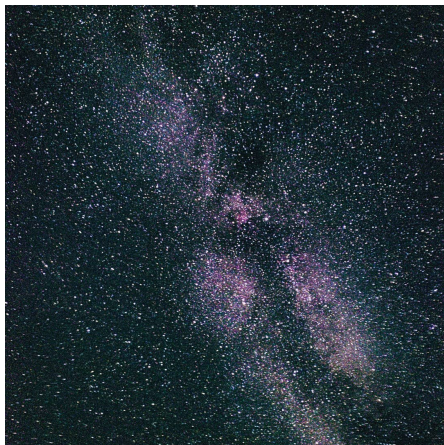
The Moon through my first telescope, the phone held to the eyepiece. Mazyr, August 2020.



The telescope that came the second New Year, the one that once showed me the neighbors' television across the yard, at the open window of my room on the fifth floor. Mazyr, June 2020.



One raw frame of the Milky Way, 32 seconds at ISO 3200, before stacking. Our country house, 10 August 2021.



Twenty-one such frames stacked into one picture. My friends asked whether it was Photoshop. August 2021.



Star trails and a meteor over the lake near my grandfather's house, stacked phone exposures. August 2021.



Fifty exposures of 32 seconds, nearly half an hour of sky, over my grandparents' house. August 2021.

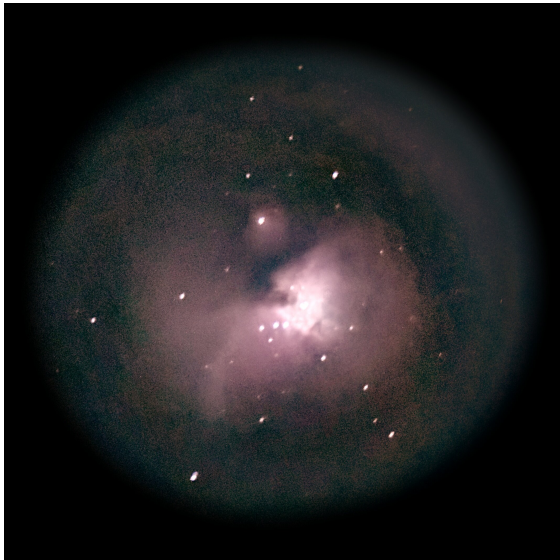
At Berea. The physics department's 8-inch telescope, an iPhone held to the eyepiece, 2024 to 2026.



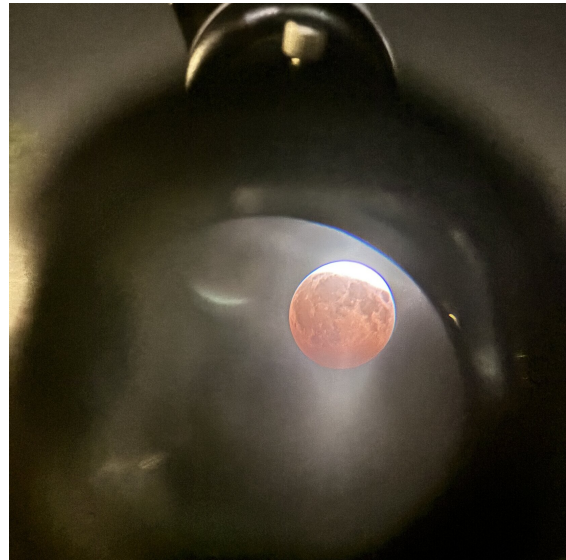
A visitor at the 8-inch on the observing deck of the science building during International Observe the Moon Night, which I helped set up. 14 September 2024.



The Moon through the 8-inch that night, the phone held to the eyepiece. 14 September 2024.



The Orion Nebula, M42, five phone frames at the eyepiece of the 8-inch, stacked, the night I first saw it, on a screen. Berea, 24 February 2025. One of the hundred Messier objects of the March marathon, so one percent done.



The partial lunar eclipse in the eyepiece during the Physics Club's public night on the alumni field. 27 August 2026.

Hosts and correspondents

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I wrote to the observatories, societies and individual observers along the route of this year, and these are the replies the year rests on, in the order I will meet the people, with one sentence of what each wrote. Words in quotation marks are theirs. The ten in the application's project contacts are marked with an asterisk. Every reply, with more of each person's words, is at melnichenka.com/eyepiece-full.pdf.

Where and when	Who	What they wrote
Egypt, 22 July to about 20 August 2027, and a March 2028 return. The Luxor eclipse of 2 August.		
Kottamia, 26 July 2027	Dr. Bassem Nabawy* President, National Research Institute of Astronomy and Geophysics, which runs the largest telescope in North Africa	"You are most welcome to Katamiya on 26 July 2027." A hosted night, and my request for a place at the institute's Luxor eclipse camp once the camp fee is set.
Luxor, 2 August 2027	Dr. Awni Al-Khasawneh* Secretary General, Arab Union for Astronomy and Space Sciences	The Union's societies observe the eclipse at Luxor under his supervision, and I may coordinate my part directly with him. "We are pleased to support your project."
Luxor, eclipse week	Prof. Zainab Awad Cairo University, co-chair of its eclipse conference	"You're most welcome to join the conference and attend the observations session."
Cairo or Luxor	Prof. Yasser Abdel-Hadi and Prof. Amir Hassan National Research Institute of Astronomy and Geophysics	Delighted to meet in Cairo or at Luxor on eclipse day and "explore possibilities for joint observations," and "prepared to contribute either personally or through a colleague" to the twilight observations at Kottamia or Luxor.
Luxor, 2 August 2027	Dr. Hisashi Hayakawa Nagoya University, who measures the Sun's radius from eclipse timings	"I'm looking forward to hearing about your measurements." Position to an arcsecond, time to a second.
Luxor, the first mornings of August	Rumen Kolev Author of The Babylonian Astrolabe	The morning Sirius first shows before dawn, which began the ancient Egyptian year, falls at Luxor between 31 July and 3 August 2027 by his tables.
Luxor, the evening of 2 August	Mohammad Odeh Director, International Astronomical Center, keeper of the world's crescent-sighting records	"A negative observation made by an experienced and reliable observer, under suitable and ideally good observing conditions, is no less important than a positive observation."
Chile, late August to October 2027.		
Elqui Valley, the new Moon of 31 August	Gabriela López* Head, Office for the Protection of the Quality of the Sky of Northern Chile	"You can count on our support to accompany us during night sky brightness measurements," made at a professional observatory.

Where and when	Who	What they wrote
Las Campanas, one night	Dr. Juna Kollmeier* Director, Sloan Digital Sky Survey	“Yes, this can be arranged.” She added that my adviser Alex Lazarian once promised her he could read the turbulent field from the survey’s data alone, “maybe you will prove that he was telling the truth!”
Valparaíso, 13 to 20 October 2027	Domingo González* President, Sociedad Astronómica de Valparaíso y Viña del Mar, an amateur society	“It will be a pleasure to receive you, share experiences and learn more about your outstanding career.”
Vicuña, August to September 2027	Juan Daniel Durán Rojas Mamalluca municipal observatory	Willing in principle to evaluate my taking part alongside the observatory’s guides and astronomers, to be coordinated at the start of 2027.
La Silla	Dr. Petr Kabáth Czech Academy of Sciences, a spectrograph at the European Southern Observatory’s La Silla	A day visit “should be possible,” and “there is no eyepiece for observing by eye during the night.”
Las Campanas	Carol Rojas Outreach, Las Campanas Observatory	Answered with the question of the whole year. “Would you rather talk to someone who knows the night sky by heart, rather than someone who works at an observatory and observes through a telescope?”
Argentina, November to December 2027.		
Bosque Alegre, 27 and 28 November 2027	Iván Bustos Fierro* Coordinator of the Bosque Alegre station, Córdoba Astronomical Observatory, National University of Córdoba	“I am quite sure you will be able to accompany them, just as our guest,” on the station’s public nights, and on staying through a run of the 1.54 meter telescope, “I am fairly certain it will be possible.”
Barreal, 29 November to 8 December 2027	Dr. Antonio Damián Alejo* Félix Aguilar Observatory, San Juan	“I would be very glad to help you arrange an opportunity for visual observing during your stay.”
Peru, January to February 2028.		
Piura, 26 January 2028	Dr. José del Carmen Silva Mechato* Dean of the Faculty of Sciences, National University of Piura	“Nos comprometemos,” we commit, to carry out the observation of the annular eclipse I proposed for its Castilla campus, with the directors of its physics and mathematics schools.
Chankillo, February 2028	Dr. Iván Ghezzi Director of Chankillo, the oldest solar observatory in the Americas	“You are welcome to visit and conduct your research,” with a formal letter first, standing a few meters in front of the western observing point.
Trujillo, 28 to 30 January 2028	Physics Student Center National University of Trujillo	“We would be absolutely delighted to welcome you!” A place at one of their meetings and a request for a talk.
Sacred Valley, 9 to 23 February 2028	Fernando Sánchez Montero Sembrando Cielos, Calca	“In principle I would be happy to have you join us for a night,” the Andean dark constellations by eye, and an indoor session if the rains close the sky.

Where and when	Who	What they wrote
Australia, March to May 2028, and the eclipse of 22 July 2028.		
New South Wales, autumn 2028	Prof. Ray Norris* CSIRO Space and Astronomy, a leading researcher of Aboriginal astronomy	Has introduced me to a Gamilaraay elder, who is happy to speak with me, and passed on his address, so that I learn the Emu in the Sky by eye.
Gamilaraay country, April 2028	Mark and Wendy Osmond Dark Sky Eco	“By eye first. No telescope. That is where the Cycle of Creation starts, and it is where the Emu lives.” At the eyepiece guests say what they never say at the screen, “That’s actually there.”
Canberra, April 2028	Canberra Astronomical Society	“We’d be delighted to show you the southern night skies during your visit, either from a dome on Mt Stromlo or from a truly dark location just outside the lights of Canberra.”
Sydney and Dubbo, 22 July 2028	Meredith Hall Inspiring Australia, and Peter Starr Dubbo Observatory	“We would be glad to assist with finding you a volunteer opportunity to be a part of the eclipse celebrations,” and a place on Dubbo’s eclipse-day list, on the central line.
New Zealand, May to mid July 2028.		
Kaikōura, 8 to 21 May 2028	Lisa Bond Kaikōura Dark Sky Trust	“You would certainly be very welcome to visit us,” a round of the sanctuary’s sky-quality readings with the team if weather and timing allow.
Wairarapa, May to June 2028	Haritina Mogoşanu Star Safari observatory and Spaceward Bound New Zealand	“More than happy to have you help with both,” her observatory and her program for teachers. “Yes you can say that.” Offers a letter.
Naseby, June to July 2028	Paul Bishop Night-sky tours, Central Otago	Almost every evening of June and July free, and a 12-inch telescope driven by hand. “That sounds perfect!”
Dunedin, 26 June to 15 July 2028	Georgette Watkins* Planetarium Officer, Tūhura Otago Museum	Volunteer places in and around the planetarium, “happy for you to mention we have opportunities available for you,” and a plan for the shows.
Dunedin, 14 July 2028	Michelle Taiaroa University of Otago, Kāi Tahu	The Matariki dawn ceremony on the museum lawn. “You will start to hear people call out names. They are the names of their loved ones that have passed away since the last Matariki.”
Advisers on the eye, for every leg.		
Anywhere	Stephen O’Meara Discovered the spokes in Saturn’s rings by eye	The confidence to catch a tenth of a magnitude “came from observations of variable stars,” and he applied it to the rings.
Anywhere	Gary Poyner British Astronomical Association, 370,000 visual estimates	“There is something very special in seeing a rare outburst of a dwarf nova, sometimes for the very first time.”
Anywhere	Dr. Mitsuru Sōma Vice President for Grazing Occultation Services, International Occultation Timing Association	“A beginner also can make a graze report. I look forward to your first graze report.”
Anywhere	Dr. Michele Bianda Specola Solare Ticinese, Locarno, the reference observatory for the world’s sunspot number	Why the sunspot number is still counted “by eye, pencil and paper sheet,” so that four centuries stay comparable.